

TUTORIAL GUIDE

Data Integration and Visualization Tool

Version 1.0

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1. Introduction

This tutorial provides a step-by-step guide to help users operate the Data Integration and

Visualization Tool. The purpose of this tutorial is to help users understand how to navigate

the system, upload datasets, generate visualizations, and manage project information.

The application is designed to integrate multiple sources of aerodynamic data into a centralized platform. Users can upload datasets, organize project records, perform analysis,

and generate visual plots for comparison and interpretation.

This tutorial is written for new users who may not have prior experience with the system.

Each section explains the basic operations of the application in a simple and structured manner.

2. Purpose of the Tutorial

The main objectives of this tutorial are:

- To guide users through the basic functions of the system
- To demonstrate how to upload and manage datasets
- To explain how to generate data visualizations
- To show how project information and records can be maintained
- To help users perform analysis using the available tools

By following this tutorial, users will be able to operate the system efficiently and perform

their required tasks without difficulty.

3. System Overview

The Data Integration and Visualization Tool is a web-based platform that allows users to manage projects, organize datasets, and generate visual representations of aerodynamic data.

The system integrates multiple data sources including:

- Computational Fluid Dynamics (CFD) data
- Wind Tunnel data
- Flight test data
- Aerodynamic datasets

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The application provides a centralized environment where users can upload files, manage records, generate analytical plots, and store reports for future reference.

4. Accessing the Application

The application can be accessed using a standard web browser.

Step 1: Open the Application

- Open a web browser such as Google Chrome, Microsoft Edge, or Firefox.
- Enter the application URL.

- The login page of the system will appear.

Step 2: Login to the System

- Enter your registered Email ID in the login field.
- Enter your Password.
- Click the Login button.
- After successful authentication, the system will open the main dashboard page.

If login credentials are incorrect, the system will display an error message and the user must

re-enter the correct credentials.

Login Page

5. Dashboard Overview

After logging into the application, the user will see the Dashboard.

The Dashboard shows a summary of projects and datasets in the system. From here, users can open projects and navigate to different modules.

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The dashboard view depends on the role assigned to the user.

5.1 User Roles

The system supports multiple user roles. Each role has different levels of access.

User Type

Roles

Access Level

Type 1

GD (Group Director), DH (Division Head) Full access to the system

Type 2

TL (Team Lead), SM (System Manager),

OIC (Officer in Charge)

Access to assigned projects

Type 3

JRF, SRF, CE, Students

Access to view and work with

project data

5.2 Group Director / Division Head Dashboard

Users with GD or DH roles can see a complete system overview.

The dashboard shows:

- Total Projects
- CFD Data Count
- Wind Data Count
- Flight Data Count
- Aero Data Count
- Total Reports

Below the summary, the list of projects is displayed.

Users can open a project by clicking View Project.

GD Dashboard

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5.3 Administrator Dashboard

The Admin dashboard is used to manage system users.

Admin can:

- View total users
- See user roles

- Create new users

To create a new user:

- Enter User Name
- Select Role
- Enter Login ID
- Enter Password
- Click Create

5.4 TL / Student Dashboard

Admin Dashboard

Users such as Team Leads, JRF, SRF, CE, and Students will see a simpler dashboard.

This dashboard mainly shows the projects assigned to the user.

To open a project:

- Find the project in the list
- Click View Project

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Student Dashboard

6. Creating and Accessing a Project

Projects are used to organize datasets and records within the system.

Steps to Create a New Project

- Open the Dashboard after logging into the system.
- Click the + New Project button located in the Projects section.

Dashboard with New Project Button

- The Add Project window will appear.

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Add Project Form

- Enter the required details:

1. Project Name - Name of the project
2. Description - Short description of the project
3. Add Members - Select users who will be part of the project

- Click the Create button to create the project.
- After the project is created, it will appear in the Projects list on the Dashboard.
- To open the project, click the View Project button.
- The system will open the Project Workspace, where users can manage project datasets and records.

Project Workspace screen

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7. Uploading and Previewing Datasets

The system allows users to upload aerodynamic datasets and preview them before analysis.

Supported dataset types include:

- o CFD Data
- o Wind Tunnel Data
- o Flight Data
- o Other data files

Steps to Upload a Dataset

- Open the Project Workspace.
- Select the required dataset tab such as:

1. CFD Data
2. Wind Tunnel Data
3. Flight Data

4. Others

Dataset Tabs in Project Workspace

- Click the Upload File button.
- The Upload Files window will appear.

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Upload Files Window

- Click Browse Plot Files and select the dataset file from your system.
- Enter a Folder / Tag Name to organize the uploaded files.
- Select the Data Type from the dropdown list.
- Click Upload.

The dataset will now be uploaded and stored in the project workspace.

Viewing Uploaded Datasets

After uploading the dataset, the files will appear inside the selected Tag / Folder.

Users can see:

- File name
- Created date
- Available actions

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Uploaded Files List

Previewing Dataset Files

After uploading datasets, users can preview the files before performing analysis.

Steps to Preview a Dataset

- Open the Project Workspace.
- In the dataset section (for example CFD Data), you will see a list of Tag/Folder names.
- Click the Go to (?) button for the required Tag/Folder.
- The system will open the File List Page, where all uploaded files inside the selected folder are displayed.

File List Screen

- In the Action column, click the View icon (?) for the file you want to preview.

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Raw Preview

When the file is opened, the system displays the Raw Preview tab.

This displays the original uploaded file including metadata.

Raw Preview Screen

Processed Preview

To view the structured dataset:

- Click the Processed tab.
- The system will display structured tabular data used for plotting and analysis.

Processed Preview Screen

Other Tabs

Inside the file view, additional tabs are available:

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- Plot - Shows generated visualization plots
- Others - Displays other related files or attachments

Users can switch between these tabs to explore the dataset further.

8. MATLAB (.MAT) File Handling

The system provides dedicated support for MATLAB (.mat) files, which contain multiple variables instead of standard tabular columns. A specialized interface is provided to

preview, select variables, and generate visualizations from .mat data.

Uploading MATLAB Files

- Open the Project Workspace
- Click Upload File
- Click Browse Plot Files
- Select a .mat file from your system
- Enter Folder / Tag Name
- Select Data Type
- Click Upload

After uploading, the file will be available in the dataset folder.

Accessing MATLAB File

- Navigate to the uploaded dataset
- Click the View (?) icon for the .mat file
- The system opens a dedicated MATLAB preview interface.

MATLAB Raw Preview

Unlike CSV or Excel files, .mat files contain multiple variables.

The Raw Preview displays:

? List of variables

? Each variable shows:

- Name
- Dimensions
- Data type
- Users can:

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? Select any variable

? View its data in tabular format

? Scroll through values

MATLAB File Raw Preview

Slice Selection

For large datasets, users can load only a portion of data.

Steps:

1. Enter slice values (example: :, 0:10, etc.)
2. Click Apply

Only the selected data range will be displayed and used.

Generating MATLAB Visualization

Select MAT Chart Type (e.g., Scatter)

Click Generate Plot

The system generates visualization based on selected variables.

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MATLAB Plot Configuration

MATLAB Calculation Module

The system allows performing calculations on .mat variables.

Steps:

- Navigate to Calculation Tab
- Select Dataset, Tag, and .mat file
- Enter formula
- Map variables
- Click Process Formula
- The system calculates results and shows preview.

- Click Save Derived Variable to store results

MATLAB File Calculation

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9. Generating Data Visualizations

The Visualization module allows users to create charts from uploaded datasets.

Steps to Generate a Plot

- Open the Project Workspace.
- Click Visualize from the left menu.

Visualization Module

- In the Plot Setting section, select the required options.
- Select the Dataset Type.
- Select the Tag and File containing the dataset.
- Choose the Plot Type:

1. 2D

2. 3D

- Select the Chart Type.
- For 2D plots, available options include:

1. Scatter

2. Line

3. Bar

4. Polar

5. Histogram

6. Box

7. Violin

8. Heatmap (X vs Y)

9. Contour

10. Scatter Line

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- For 3D plots, available options include:

1. 3D Scatter

2. 3D Line

3. 3D Surface

- Select the X-Axis and Y-Axis parameters.

- For 3D plots, also select the Z-Axis.

- Choose the Scale type for the axes.

- Available scale options include:

1. Linear : This is normal scale used in most graphs, in this values increase evenly on the axis.

2. Log : In this scale, values increase exponentially instead of evenly, Both X and Y axes use logarithmic scaling.

3. Semi-Log : In this scale, only one axis (either X or Y) is logarithmic, while the other remains linear. This is useful when one variable has a large variation and the other does not.

- Optionally enter a Plot Name.

- Click Generate Plot.

Contour Plot

Contour plots are used when you need to analyze relationships between three variables.

Required Inputs:

- X-axis ? Independent variable (Example: Alpha)

- Y-axis ? Independent variable (Example: Beta)
- Z-axis ? Dependent variable (Example: Lift Coefficient - CL)

How System Handles Missing Data:

- If Z values are not directly available:
- The system automatically computes them using interpolation
- This ensures smooth and continuous contour visualization

The system will process the selected dataset and display the generated visualization.

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Generated 2D Plot

Generated 3D Plot

10. Creating Multiple Plots (Over-Plot)

The system allows users to compare multiple datasets on the same visualization.

Steps

- Generate a plot using the steps above.
- Click the + Over Plot button.

Over Plot Button

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- Select another dataset, file, and axis parameters.
- Click Generate Plot again.
- The system will add the new dataset to the existing visualization for comparison.
- Overplot is typically used to compare CFD vs Wind Tunnel vs Flight data.

Multiple Plots

11. Performing Calculations

The Calculation feature allows users to create a new column by applying a mathematical formula to existing dataset columns.

This helps users generate additional values that can be used for analysis and visualization.

Steps to Perform a Calculation

- Open the Project Workspace.
- Click Visualize from the left menu.
- Select the Calculation tab at the top of the page.

Calculation Module

- Select the Dataset from the dropdown list.
- Available options include:

? CFD

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? Wind Tunnel

? Flight Data

? Others

- Select the Tag that contains the dataset.
- Select the File that you want to use for the calculation.

Create a Derived Column

- Enter a name in the Derived Column field.
- This will be the name of the new column that will store the calculated values.

Enter the Formula

- In the Formula field, type the mathematical expression you want to apply.
 - Example: $\sqrt{a} + b$
 - While typing the formula, the system may show suggestions for available functions
- You can select a function from the suggestion list.

Formula Input and Suggestions

Map Variables to Dataset Columns

- After entering the formula, the system detects the variables used in the formula.
- Select the corresponding dataset column for each variable.
- Click Process Formula.
- The system will calculate the values using the provided formula and display a preview table showing the results.
- Save the Derived Column
- After verifying the preview results, click Save Derived Column.

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- The new calculated column will be saved and added to the dataset.

Calculation Preview

12. Management Dashboard

The Management Dashboard serves as a centralized platform for monitoring, organizing, and governing all project-related and organizational records across the system.

For senior roles such as Group Director (GD) and Department Head (DH), the dashboard provides a comprehensive, cross-project view of all records, enabling:

- Centralized Visibility: Access to records across all projects, departments, and activities from a single interface.
- Decision Support: Quick review of key documents such as Minutes of Meetings, Technical Reports, Budget Estimates, and Procurement Records to support informed decision-making.
- Progress Tracking: Monitor project activities, team engagements, and record updates in real time.
- Governance & Oversight: Ensure compliance, standardization, and proper documentation across all projects.
- Efficient Retrieval: Easily search, filter, and access historical and current records without navigating multiple systems.

The system provides several modules to store and manage important project documents and records. These modules help users upload, organize, and track project-related

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information in one place. The record management modules are available in the left navigation menu.

- Available Record Modules are :

1. Minutes of the Meeting
2. Student Engagement
3. Procurement Records
4. Budget Estimation
5. Divisional Records
6. Customer Feedback
7. Training Records
8. Technical Reports
9. Digital Library

- Users can open any module by clicking the corresponding option from the navigation menu.

Project Modules

Uploading a Record

Users can upload documents such as reports, meeting minutes, or project files.

Steps to Upload a Record

- Open the required module from the left navigation menu
- Click the Upload Files or Upload button.
- The Upload Document window will appear.

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- Click Browse File and select the document from your computer.
- Supported file formats include PDF and Word documents.
- Enter the required details
- Click Upload.
- The document will be stored in the selected module and will appear in the file list.

Viewing Records

- Uploaded records are displayed in a table format showing:

1. File Name
2. Type
3. File Size
4. Created Date

- Users can open a document by clicking the View icon.

Editing Record Information

- If changes are required, users can update the record details.

Steps

- Locate the required record from the list.
- Click the Edit icon.
- Modify the required information.
- Click Save Changes.

Downloading Records

Users can download stored documents for reference.

Steps

- Locate the required record.
- Click the Download icon.
- The file will be downloaded to the user's computer.

Deleting Records

If a record is no longer required, it can be removed from the system.

Steps

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- Locate the record in the list.
- Click the Delete icon.
- Confirm the deletion.

13. Notifications

- The system includes a notification feature to inform users about system activities.
- Users can view notifications by clicking the notification icon located in the top-right corner of the interface.

Notification Section

14. Changing the Account Password

Users can update their account password using the Settings option available in the system.

This helps keep the account secure and prevents unauthorized access.

Steps to Change the Password

- Open the Settings option from the left navigation menu.

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- The Reset Password section will appear.
- Enter your Current Password in the first field.
- Enter a New Password in the next field.
- The new password must contain at least 8 characters and one special character.
- Re-enter the New Password in the Re-Enter New Password field to confirm it.
- Click the Save Password button.
- After clicking Save Password, the system will update the password successfully.

Password Reset Page

15. Conclusion

The Data Integration and Visualization Tool provides a structured and centralized environment for managing aerodynamic datasets and generating analytical visualizations. The system enables integrated analysis of CFD, Wind Tunnel, and Flight data, supporting advanced visualization, comparison, and decision-making for aerodynamic studies. By following the steps described in this tutorial, users can easily navigate the system, upload datasets, perform data analysis, generate visualizations, and manage project information efficiently.

The platform supports key capabilities such as multi-source data integration, over-plot comparison, derived calculations, and advanced visualization techniques including 2D, 3D, contour, and log-scale plots.

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This tutorial ensures that new users can quickly learn how to operate the application and effectively utilize its features for aerodynamic data analysis and engineering insights.

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